

National Exit Examination Blueprint (SAMPLE)

Program: BSc in Computer Science

Total exam items: 100 Total credit hours of examinable courses: 28

Theme	Course	Cr. Hr	Items	Weight %
1. Programming (35% of items)	Computer Programming	4	10	10
	Object Oriented Programming	4	10	10
	Data Structures and Algorithms	4	15	15
2. Systems (40% of items)	Operating Systems	4	15	15
	Database Systems	4	15	15
	Computer Networks	3	10	10
3. Theory (25% of items)	Automata and Complexity Theory	3	10	10
	Compiler Design	2	15	15

Theme credit hours: Programming 12, Systems 11, Theory 5.

Learning outcomes examined per course

Data Structures and Algorithms

- Implement stacks, queues, linked lists and trees
- Analyse algorithms with Big-O notation
- Apply sorting, searching and graph traversal

Operating Systems

- Explain process scheduling and synchronization
- Detect and avoid deadlock
- Compare paging and segmentation

Database Systems

- Design ER models and map them to relations
- Normalize schemas up to BCNF
- Write SQL queries with joins and aggregation

Compiler Design

- Describe the phases of a compiler
- Build lexical analysers from regular expressions
- Construct LL(1) and LR parsing tables